Credit Name: CSE3010 - Object-Oriented Programming 2

Assignment Name: UEmployee, Faculty, Staff

How has your program changed from planning to coding to now? Please explain?

**Understanding The Problem**

I understood the assignment as creating a program that allowed the user to input a first name, last name, salary and whether the employee was Staff or Faculty. If the employee was a staff member, the program would ask the user for a job title; if the employee was a faculty member, the program would ask the user for a department. The UEmployee would be the superclass and Staff and Faculty would be the subclass. Both Staff and Faculty would inherit the employee's first name, last name and salary beforing asking for an input specific to the subclass. The instructions were clear.

**Planning The Solution/Implementation**

I broke it into smaller steps, starting with the superclass(UEmployee). I created the constructor method, overloading method, modifier method, access method and a toString method that returns the employee's information. I then created the Faculty subclass the same way, using *super* to inherit the values from the superclass and adding department. I repeated the same steps for the Staff subclass. In the client class, I started with a method to return an object; I prompted to user to enter a first name, last name and salary before asking whether the employee was a staff or faculty in a while loop. I used the loop to ensure that a valid input was entered to ensure that an object would be returned. Based on whether the employee was staff or faculty, I prompted the user to input a job title or department, then returning an object with the information inputted. I used a do-while loop to continue running the program until "e" was entered. I created a arraylist for the objects and added each object returned to it in order to organize the objects. I then prompted the user for an action: continue assigning employees, view employees or end program. If the user wanted to assign more employees, the loop would begin again and prompt the user for information. If the user wanted to view the employees, I used a for-loop to to print each object in the arraylist; due to the having a toString method, I am able to print a String that represents the object. I also added a while loop for when the user enters an invalid value, prompting the user until a valid input is entered. When things did not display correctly due to the while loops, I used *System.out.print* to figure out where it went wrong with the use of trial and error.

**Overcoming Challenges**

The most difficult part was deciding how to display the program. The problem did not ask for the program to be display a certain way, allowing me creative freedom. Going over the CresMart skillbuilder provided an example, helping me get an idea to what it should look like.

**Learning**

This program provided a basic example of what subclasses should look like, providing a base that I can look back to in the future.